

Website Traffic Analysis

Executive Summary

This project analyzes website traffic data to identify visitor patterns, peak activity periods, and overall website performance. The dataset contains 2,793 traffic observations recorded between January and May 2020.

The analysis revealed that website traffic is highly concentrated around specific periods, with several high-traffic events generating a significant portion of total visits. Traffic activity is strongest during weekday business hours, particularly between 10 AM and 2 PM, while weekends experience lower engagement levels.

The findings provide valuable insights for scheduling marketing campaigns, optimizing content publication timing, and improving website infrastructure planning.

Dataset Overview

- Total Records: 2,793
- Features: 2
 - Timestamp
 - TrafficCount
- Missing Values: 0
- Tools Used:
 - Python
 - Pandas
 - Matplotlib
 - Seaborn
 - Jupyter Notebook

Website Traffic Analysis – Visualizations & Insights

```
[5]: import os

print(os.listdir(r"D:\DA INTERN\da vs"))

['Customer_Behavior_Analysis', 'Sales performance analysis', 'web_traffic.csv']
```

```
[6]: import os

for root, dirs, files in os.walk(r"D:\DA INTERN\da vs"):
    for file in files:
        if file.endswith(".csv"):
            print(os.path.join(root, file))

D:\DA INTERN\da vs\web_traffic.csv
D:\DA INTERN\da vs\Customer_Behavior_Analysis\eCommerce_customer_data_custom_ratios.csv
D:\DA INTERN\da vs\Customer_Behavior_Analysis\eCommerce_customer_data_large.csv
D:\DA INTERN\da vs\Sales performance analysis\SuperStoreOrders.csv
```

```
[7]: import pandas as pd

df = pd.read_csv(r"D:\DA INTERN\da vs\web_traffic.csv")

df.head()
```

```
[7]:
```

	Timestamp	TrafficCount
0	2020-01-20 00:00:00	487.0
1	2020-01-20 00:30:00	385.0
2	2020-01-20 01:30:00	269.0
3	2020-01-20 02:30:00	517.0
4	2020-01-20 03:30:00	129.0

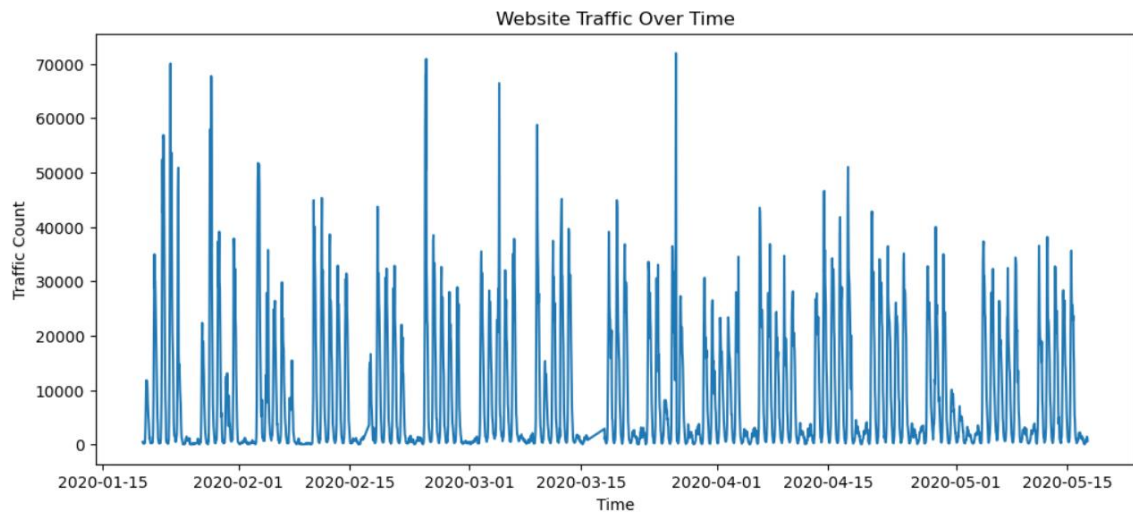
1. Website Traffic Over Time

```
[8]: plt.figure(figsize=(12,5))

plt.plot(df['Timestamp'], df['TrafficCount'])

plt.title("Website Traffic Over Time")
plt.xlabel("Time")
plt.ylabel("Traffic Count")

plt.show()
```



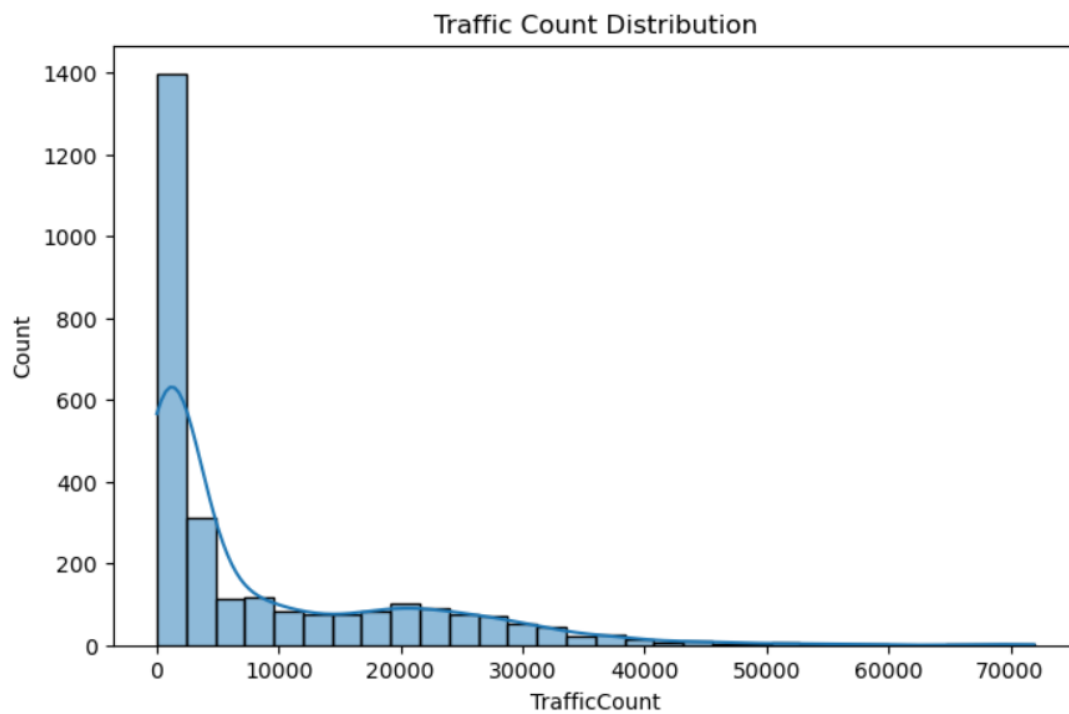
2. Traffic Count Distribution

```
[ ]: plt.figure(figsize=(8,5))

sns.histplot(df['TrafficCount'], bins=30, kde=True)

plt.title("Traffic Count Distribution")

plt.show()
```



3. Average Traffic by Day

```
13]: daily = df.groupby('Day')['TrafficCount'].mean()

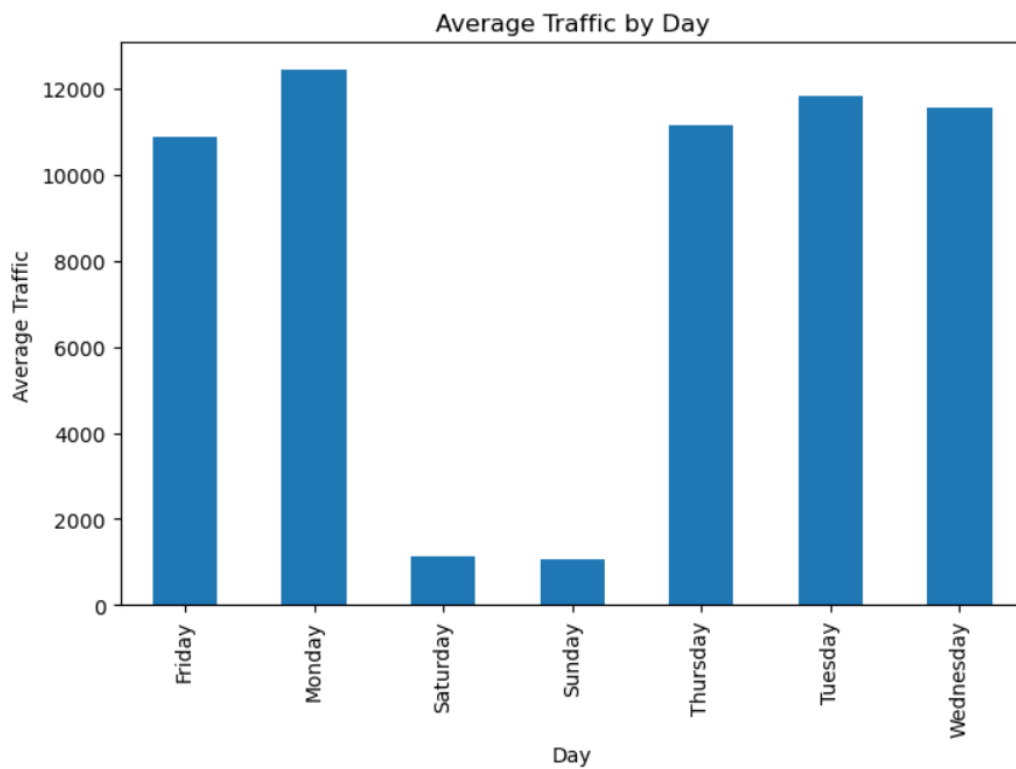
plt.figure(figsize=(8,5))

daily.plot(kind='bar')

plt.title("Average Traffic by Day")

plt.ylabel("Average Traffic")

plt.show()
```



4. Average Traffic by Hour

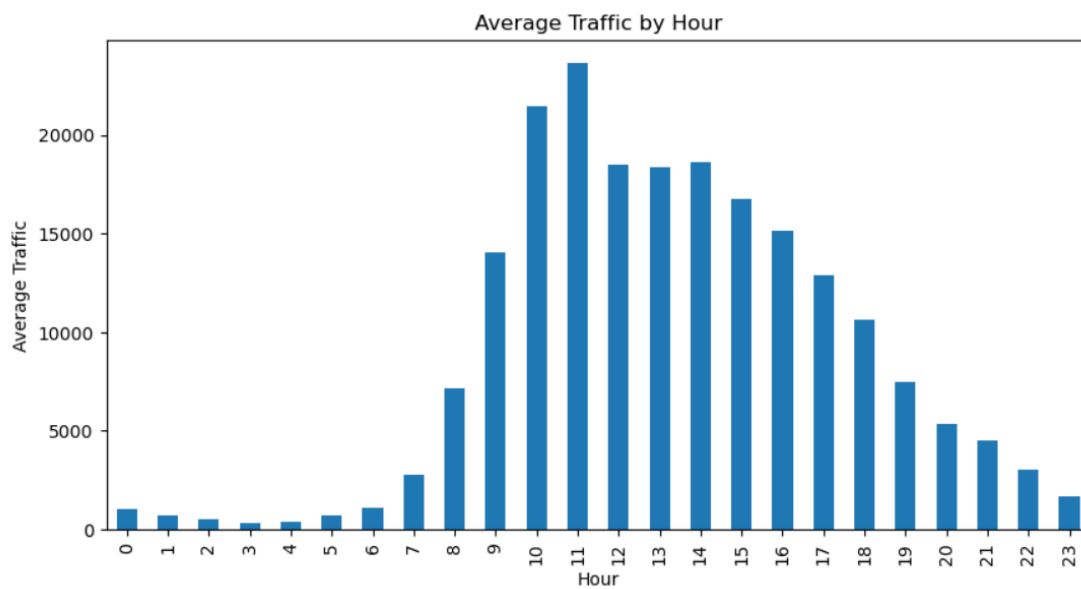
```
: hourly = df.groupby('Hour')['TrafficCount'].mean()

plt.figure(figsize=(10,5))

hourly.plot(kind='bar')

plt.title("Average Traffic by Hour")
plt.ylabel("Average Traffic")

plt.show()
```



5. Top 10 Traffic Days

```

|: daily_total = df.groupby('Date')['TrafficCount'].sum()

top_days = daily_total.sort_values(ascending=False).head(10)

plt.figure(figsize=(12,5))

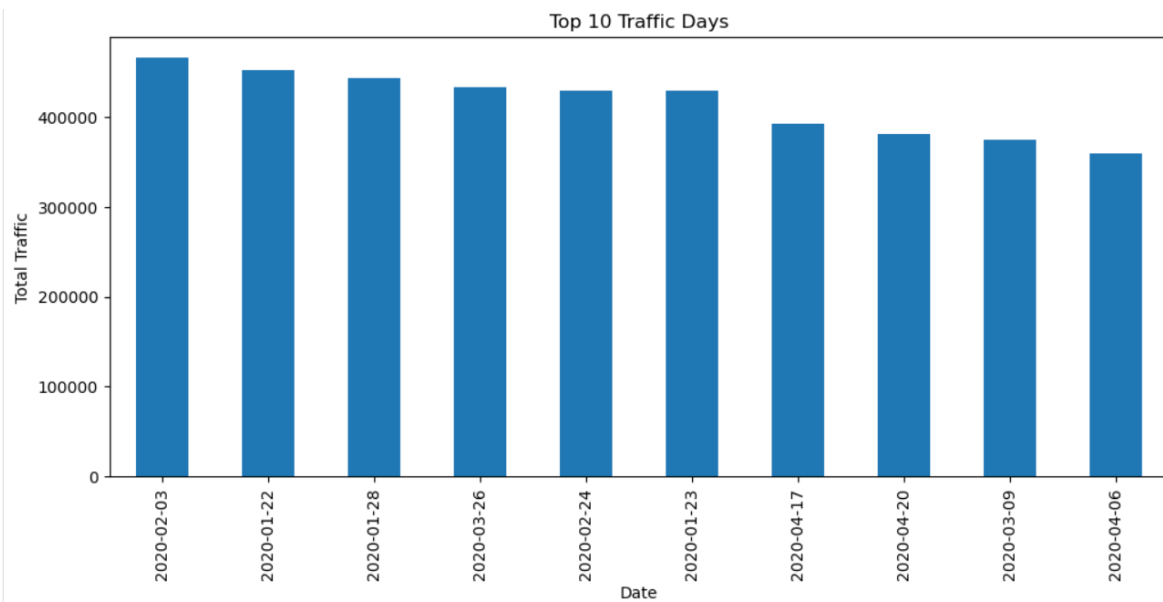
top_days.plot(kind='bar')

plt.title("Top 10 Traffic Days")

plt.ylabel("Total Traffic")

plt.show()

```



Key Findings

Traffic Statistics

- Maximum Traffic: 71,925
- Minimum Traffic: 22
- Average Traffic: 8,591
- Median Traffic: 2,418

Hourly Traffic Insights

- Traffic begins increasing around 8 AM.
- Peak traffic occurs between 10 AM and 2 PM.
- Highest traffic observed around 11 AM.

Weekly Traffic Insights

- Monday generated the highest traffic.
- Weekdays consistently outperformed weekends.
- Saturday and Sunday recorded the lowest traffic.

Traffic Pattern Analysis

- Traffic distribution is highly skewed.
- A small number of high-traffic events generated significant website activity.
- Several traffic spikes were observed throughout the analysis period.

Business Recommendations

1. Schedule Marketing During Peak Hours

Run campaigns between 10 AM and 2 PM to maximize engagement.

2. Focus on Weekday Promotions

Allocate advertising budgets primarily to weekdays.

3. Analyze High-Traffic Events

Investigate successful traffic spikes and replicate effective strategies.

4. Improve Infrastructure Planning

Prepare servers and resources for predictable traffic surges.

5. Implement Continuous Monitoring

Track website performance and visitor behavior to identify future growth opportunities.

GITHUB LINK:- <https://github.com/TejasSinha2006/Website-Traffic-Analysis>